AvatarForcing: One-Step Streaming Talking Avatars via Local-Future Sliding-Window Denoising

Supplementary Material

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Overview. This supplementary material provides: (1) additional implementation details of AvatarForcing, including heterogeneous noise scheduling, streaming DMD distillation, and KV-cache updates; (2) training configuration and runtime measurements, including per-module latency breakdown and the end-to-end delay protocol; (3) construction details of the long-form benchmark and the exact evaluation pipeline for long-horizon stability metrics (ΔE_{2000} and Adj-LPIPS); (4) additional ablations and qualitative analyses; (5) user study protocol/results; and (6) failure cases and challenging scenarios.

1 Implementation and Evaluation Details

1.1 Heterogeneous Noise Scheduling

Algorithm 1 Streaming DMD Distillation with Rolling Window and RoPE Re-indexing

Require: Sliding-window latents $\{\mathbf{z}_w\}_{w=1}^{W}$;
Multimodal conditions $\{\text{cond}_w\}_{w=1}^{W}$;
Noise steps $\{\mathbf{t}_w\}_{w=1}^{W}$; timestep shift α; KV caches KV
for $w = 1$ **to** W **do** ▷ Emit → Slide → Append → Update
  $\mathbf{t}'_w \leftarrow \textsc{NormalizeAndShiftTime}(\mathbf{t}_w, \alpha)$
  $\hat{\mathbf{v}}^{(S)}_{w,1:B} \leftarrow f_S(\mathbf{z}_w, \text{cond}_w, \mathbf{t}'_w, \text{KV})$
  $\hat{\mathbf{v}}^{(T)}_{w,1:B} \leftarrow f_T(\mathbf{z}_w, \text{cond}_w, \mathbf{t}'_w)$ ▷ Teacher is global, no cache
  $\mathcal{L}_{\text{DMD}} \mathrel{+}= \sum_{j=1}^{B} \lambda(t'_{w,j}) \cdot \left\| \hat{\mathbf{v}}^{(S)}_{w,j} - \hat{\mathbf{v}}^{(T)}_{w,j} \right\|_2^2$
  Emit: commit the leftmost clean block to output
  Slide: shift window left
  Append: add a pure-noise block to the right → build $\mathbf{z}_{w+1}$
  Update KV: keep $\text{KV}^{\text{anchor}}$ persistent and write clean block to KV; apply RoPE re-indexing to the anchor cache
end for
return $\mathcal{L}_{\text{DMD}}$

We adopt a rolling window of L blocks, each comprising B latent frames, with heterogeneous noise levels specified by the `denoising_step_list` configuration

(default: $L = 4$, $B = 4$). All timesteps $t \in [0, 1]$ are further transformed by the `timestep_shift` mapping, which expands early denoising steps and compresses later ones, thereby improving temporal smoothness at window boundaries. This transformation is applied consistently during both training and inference.

1.2 Streaming DMD Distillation

We provide pseudocode for the streaming rollout and KV-cache update used in our two-stage distillation.

Algorithm 2 Emit-Slide-Append-Update

```
for each window do
    Emit: Generate window output
    Slide: Add noise to output for the next window
    Append: Concatenate fresh noise to the right
    Update: Refresh KV cache
end for
```

Algorithm 1 outlines the streaming DMD distillation procedure with rolling windows and RoPE re-indexing. Algorithm 2 further illustrates the corresponding KV-cache update process.

1.3 Training Configuration and Inference Latency

We use a learning rate of 1.5×10^{-6}, a critic learning rate of 4×10^{-7}, and a weight decay of 0.01. An EMA with decay rate 0.99 is applied starting from step 200. The KV cache consists of a style anchor (one frame) and a temporal anchor (four frames). All input latents and reference frames are resized to 832×480. Training is performed on 16 GPUs for 72 hours. Steady-state per-frame latency is reported in Tab. 2, and the end-to-end delay protocol is summarized in main_paper. Tab. 2.

1.4 Ablation Study on the Effect of Timestep Shift

We conduct an ablation study to assess the effect of the `timestep_shift` parameter on long-form streaming video generation. As shown in Fig. 1, we compare shift values of 1, 2, 4, and 6. Smaller shifts (1 or 2) yield a more uniform denoising schedule, which leads to unstable fine details, drifting illumination, and noticeable temporal artifacts at window boundaries. In contrast, larger shifts (4 or 6) concentrate denoising steps in the high-noise regime, effectively suppressing local artifacts and producing smoother temporal transitions, better audio-visual synchrony, and more stable details over time. These results indicate that allocating more denoising capacity to the high-noise phase is crucial for robust temporal coherence and artifact-free streaming generation.

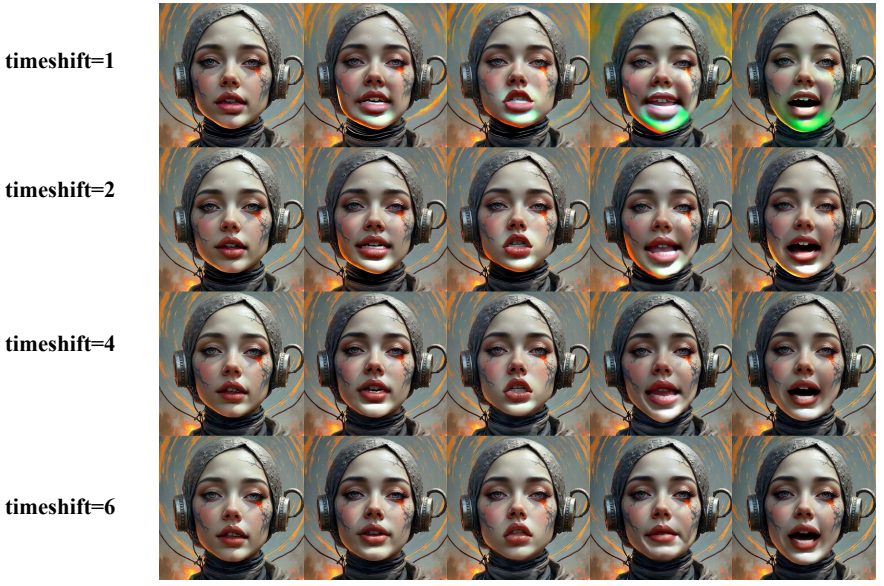

Fig. 1: Timestep-shift ablation. Larger shift values (bottom) reduce detail drift and improve temporal smoothness, while smaller values (top) result in visible artifacts and instability at window boundaries.

2 Benchmark Construction

We provide additional details of the long-form benchmark used in the main paper. All clips are center-cropped and standardized to 480p to ensure consistent, human-centric framing.

The benchmark contains nearly 400 videos ranging from 40 seconds to 2 minutes. It covers multilingual speech, singing, and instruction-following scenarios, deliberately curated to stress-test models on long-horizon robustness, identity preservation, and audio-visual alignment. It includes a wide spectrum of speaking styles, expressive motions, and head poses, as well as challenging cases such as rapid speech, occlusions, and profile views.

All samples are manually screened to remove identity ambiguities and visual artifacts, resulting in a high-quality yet challenging benchmark for evaluating streaming avatar models under realistic, extended settings, substantially beyond the coverage of conventional short-form datasets.

2.1 Metric Implementation Details for Long-Form Stability

This section details the exact evaluation protocol used to compute the long-form stability metrics reported in the main paper (ΔE_{2000} and Adj-LPIPS). We use *aligned face crops* in a two-level sense: (i) **benchmark-level alignment** via center-cropping and resizing to 480p to obtain consistent, face-centric frames, and (ii) **frame-level alignment** via landmark-based normalization when a

canonical pixelwise correspondence is required. For each video, we treat the first frame as a reference and process the remaining frames sequentially (default stride = 1). For the long-horizon tables in the main paper that report **Total** / **Last** over 2-minute rollouts, **Total** denotes the mean over the full rollout, while **Last** denotes the mean over the final 60 seconds of the same rollout (both averaged across videos). We detect the largest face in each frame using InsightFace and use its keypoints to obtain a canonical aligned crop for metrics that operate on aligned face pixels.

Color drift (ΔE_{2000}). We compute CIEDE2000 between the reference and each frame on an aligned 112×112 face crop (`norm_crop`). Both crops are converted to CIE Lab, and a per-pixel ΔE_{2000} map is computed w.r.t. the reference. To focus on skin-tone drift (rather than background/hair), we build a fixed binary skin mask from the reference crop using a YCrCb threshold $[0, 133, 77]$-$[255, 173, 127]$ followed by a 3×3 morphological open/close; if the mask is degenerate, we fall back to a central rectangular region. The per-frame score is the mean of the ΔE_{2000} map over the reference skin mask. We report the per-video mean averaged across videos.

Flicker (Adj-LPIPS). We measure flicker as the LPIPS distance (VGG backbone) between adjacent frames $(t-1, t)$ on the benchmark-aligned face-centric frames (center-cropped and resized to 480p), without additional landmark warping. Adj-LPIPS is computed for all adjacent pairs and summarized by the per-video mean averaged across videos.

3 Additional Ablations and Qualitative Analyses

3.1 Limitations of Existing Open-Source Long-Form Avatars

To contextualize the long-form benchmark, we visualize representative failure modes of publicly available long-form avatar baselines under the same evaluation protocol (center-cropped and standardized to 480p, identical audio and reference-frame inputs). Fig. 2 highlights that even methods that appear stable on short clips can accumulate errors over long rollouts, with progressive color drift and degraded lip motion becoming increasingly noticeable as the sequence length grows.

3.2 Effect of Window Length

We further provide qualitative examples of how the streaming window length L affects long-horizon stability under the same block emission setting. Increasing L introduces bounded local-future context within the window and reduces boundary artifacts, while L=1 collapses to a strictly causal update that is more prone to flicker. This qualitative trend is consistent with the quantitative sweep in the main paper (main_paper. Tab. 3).

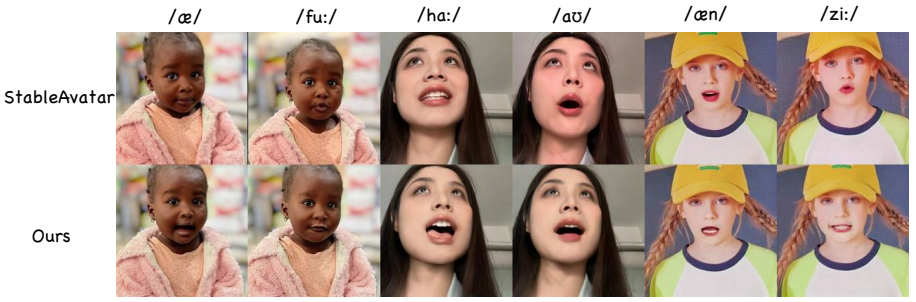

Fig. 2: Long-form failure modes of open-source baselines. StableAvatar preserves identity but exhibits progressive color shifts and inaccurate lip motion over long sequences, motivating explicit long-horizon anchoring and windowed refinement for stable streaming generation.

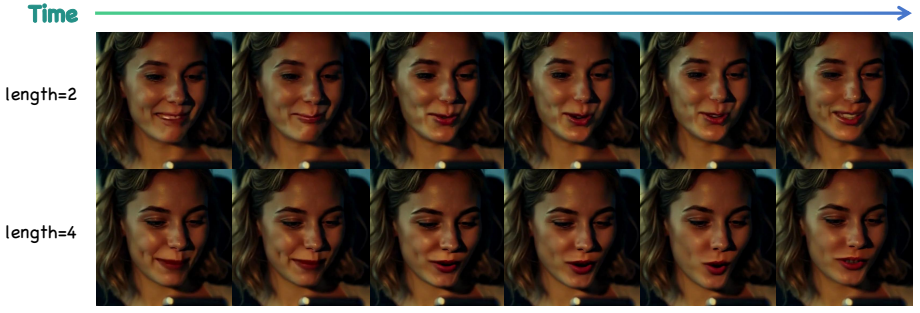

Fig. 3: Effect of window length L. A window size of 2 still shows instability; 4 yields stable long-horizon consistency by enabling effective bidirectional refinement within our heterogeneous moving buffer.

3.3 Teacher CFG Ablation

The teacher uses classifier-free guidance (CFG) during trajectory generation, while the distilled student runs without CFG at inference. Tab. 1 ablates the teacher CFG scale and shows that moderate guidance yields the best student quality and lip synchronization.

Table 1: Teacher CFG ablation.

CFG	CSIM↑	Sync-C↑	Sync-D↓
1.5	0.901	4.79	9.34
3.0	0.910	6.21	8.33
4.5	**0.911**	**6.59**	8.18
6.0	0.897	6.59	8.04

Table 2: Per-module latency breakdown.

Component	Latency(ms)	Description
Audio Encoder	3	Wav2Vec feature enc.
Video Transformer	21	Windowed DiT forward
KV Cache Update	2	KV update & re-index
VAE Decoder	8	VAE decoding
Total	34	Steady-state latency

3.4 Notes on Core Controls

The main paper includes two complementary controlled studies to attribute the gains of AvatarForcing. First, the (L, N) sweep in main_paper. Tab. 3 keeps the

same anchors and caching while varying only the window length L (look-ahead) and the number of denoising passes N (refinement), where L=1 provides a no-look-ahead reference. Second, the anchor ablations in main_paper. Tab. 4 keep the streaming mechanism fixed and isolate the role of the style and temporal anchors in long-horizon stability.

We further clarify that heterogeneous timestep ordering within the window is adopted to enable one-step streaming emission: it establishes an emit-ready clean frontier for emission while preserving noisy future blocks for correction. A homogeneous shared-timestep window removes this separation and often requires additional denoising passes before a block becomes emit-ready; therefore, it is not directly comparable as a drop-in one-step ablation under the same streaming constraint. In practice, its closest comparable endpoints are covered by the strictly causal one-step baselines (main_paper. Tab. 5) and the offline bidirectional teacher (main_paper. Tab. 4).

4 User Study

4.1 Evaluation Protocol

We conduct a user study on four long-form (>1 min) talking-avatar videos per method, each center-cropped to 512×512 for a consistent, human-centric view. All methods are evaluated with their official checkpoints and configurations under identical audio and reference-frame inputs. From the resulting videos, we randomly sample $N = 30$ anonymized and shuffled clips per method for evaluation.

Each clip is rated by 20 participants on a five-point Likert scale (1-5) according to: **Lip Synchronization Accuracy** (alignment between mouth motion and audio); **Identity Preservation** (consistency of facial features and subject identity); **Timing Stability** (temporal smoothness and absence of jitter or drift); **Image Quality** (visual sharpness and clarity); and **Overall Realism** (perceived naturalness and believability). We report mean MOS.

Table 3: User study (MOS) results. Mean opinion scores (1-5; higher is better) from 20 participants across five criteria.

Method	Lip synchronization	Identity	Timing	Image	Realism
EchoMimic	2.4	2.7	3.1	3.6	3.5
Hallo3	3.5	3.8	3.2	3.7	3.6
HunyuanAvatar	3.7	4.0	3.4	3.9	3.8
FantasyTalking	3.6	3.9	3.3	3.8	3.7
OmniAvatar	3.9	4.1	3.8	4.1	4.0
MultiTalk	3.8	4.0	3.6	4.0	3.9
StableAvatar	3.6	4.3	3.5	4.0	3.8
Ours	**4.6**	**4.5**	**4.5**	**4.3**	**4.4**

4.2 User Study Results

As shown in Tab. 3, our method achieves the highest MOS across all subjective criteria, with pronounced gains in lip synchronization accuracy, identity preservation, and overall realism over all baselines. These trends are consistent with the corresponding objective metrics. The advantages are particularly salient in long-form settings, underscoring the effectiveness of the streaming generation pipeline under challenging evaluation conditions.

5 Latency Breakdown

We report a component-wise breakdown of the steady-state per-frame latency used in the main paper (main_paper. Tab. 2) following recent protocols [4]. All timings are measured on a single GPU with batch size 1 at an output resolution of 832×480 using bfloat16 inference.

Terminology. Tab. 4 aligns commonly reported runtime terms (e.g., FPS/RTF and overall delay) with the quantities defined and reported in the main paper (main_paper. Tab. 2).

Table 4: Terminology alignment for streaming runtime metrics. We separate steady-state per-frame cost from user-visible delay and relate the main-paper definitions (main_paper. Tab. 2) to commonly reported throughput and latency terms [1,3].

Quantity (ours)	What it measures	Related terms
Steady-state latency (s/frame)	Per-frame wall-clock compute time in steady state (audio encoder + DiT + KV update + VAE), excluding initialization and I/O (main_paper. Tab. 2); throughput is its reciprocal.	FPS, RTF
Audio look-ahead T_{LA}	Future audio required by bounded local-future denoising: $(L-1)B$ frames, i.e., $(L-1)B/25$ seconds at 25 FPS (main_paper. Tab. 2).	Buffering / chunking
End-to-end delay (audio→visual)	In a real-time streaming simulation, the wall-clock delay between receiving the audio embedding for frame t and emitting the aligned video frame t; includes look-ahead buffering and steady-state compute (main_paper. Tab. 2).	Overall delay

Notes. The audio encoder and video transformer can be parallelized in batched settings for higher throughput. All latency values exclude model initialization, data loading, disk I/O, and post-processing. The total latency is comparable to recent streaming diffusion models [2, 4], and remains stable for long-form sequences due to efficient cache management and windowed inference. Tab. 2 summarizes the per-frame inference latency breakdown.

Delay protocol. We simulate real-time streaming at 25 FPS. For each frame index t, we record the wall-clock time when its audio embedding a_t becomes available to the generator. We also record the wall-clock time when the aligned video frame t is emitted to the output stream (after VAE decoding). In our

implementation, frames are emitted at a fixed 25 FPS while the model internally processes blocks of B frames per step; all timings are normalized per frame (main_paper. Tab. 2). We measure delay in steady state after the initial $(L-1)B$ frames buffer-fill period. With fixed L and near-constant per-step compute, the steady-state delay is dominated by the deterministic look-ahead buffer T_{LA} plus the steady-state per-frame compute on the critical path.

Fig. 4: Long-horizon failure cases. Representative long-horizon failure cases. As sequence length increases, accumulated pose and translation errors reduce motion amplitude and lead to subtle spatial drift (top and middle rows). Under more dynamic motion or expressive behaviors, the model further exhibits color instability, identity blur, and weakened temporal coherence (bottom row), reflecting the limits of short-window conditioning in very long-form synthesis.

6 Failure Cases and Challenging Scenarios

While our streaming pipeline achieves robust long-form generation, several limitations persist under extreme conditions.

Diminished Motion Amplitude. Beyond the 2-minute benchmark range, the motion amplitude and expressiveness of the subject can gradually fade. In longer talking segments, head and facial movements become more restricted compared to ground truth or short-range generations. This likely stems from the bounded temporal context at inference being narrower than the long-horizon dependencies implicitly learned during training.

Long-horizon Degradation. For very long videos (>5 minutes), artifacts such as subtle color shifts, loss of facial detail, and mild temporal drift occasionally appear (see Fig. 4). These artifacts become more pronounced with increasing duration and are further exacerbated by challenging conditions, for example, rapid speech or large pose changes. This degradation is attributed to residual

distribution mismatch and insufficient global context, which could be alleviated in future work by integrating stronger memory mechanisms or periodic anchor-frame resets.

Notably, a gradual spatial drift behavior is observed, where the generated avatar slowly shifts toward the image boundary over long sequences. This behavior is also reported in self-forcing models. Although a style anchor provides a persistent reference, the corrective influence is inherently limited under bounded-window streaming inference. When each new window relies primarily on short-term history, local pose and translation errors can outweigh the global signal of the anchor and accumulate at sub-pixel scales, eventually pushing the avatar off-center. The effective temporal reach of the anchor is further diluted by windowed attention and cached representations, which prevents minute-level recentering. We observed that occasionally resetting part of the clean-block KV cache can mitigate this drift, suggesting a potential direction for more robust long-range memory management.

Discussion. These failure cases highlight current limitations in modeling global coherence and long-range motion diversity in infinite-length video generation. Addressing such challenges is an important direction for future research.

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